

SCIENCE - Can, May, and the First Weld

Why Gate exists, in plain language first. Then the science labels.

Source: *gate/SCIENCE.md* | Undefined word? Open doc 18 (Layer 0 atoms) and define it before continuing.

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Plain opener (middle-school test)

Computers can do more dangerous things every year - send money, bind coverage, let an agent click a tool. Being able to do it is not the same as being allowed. Gate is a mouth on that gap: before an irreversible change finishes, check permission. If unsure, stop (DENY). Keep a receipt someone else can open.

First real job we prove: payout / withdraw - clear before the wire. Prove a real DENY on one money-leave path before talking grid or force.

What we are not

- We do not own nukes, military command, the power grid, or link monopolies.
- We contribute as a clearance mouth under people who already own those stacks.
- We do not sell a pretty dashboard that never sits on the write.

Words you must own (from 18)

- can / may
- irreversible write
- DENY / fail-closed
- LIVE / DEAD / CHARGE
- clearance vs execution

Science labels -> plain Gate law

PRI (pre-irreversibility inhibition): Do not complete an unjustified irreversible act. Execute only if LIVE and justified; else DENY. No bypass. Only CHARGE re-opens.

Fail-closed: Uncertainty means DENY. Soft-yes under panic is the failure mode.

Measure -> control -> receipt: You cannot magically 'just intervene' with zero cost. Check, decide LIVE/DENY, leave stranger-checkable evidence. Silent free yes is illegal.

Driver node (not hub dashboard): Sit on the actual irreversible edge (the write). Owning a busy overview screen is not the same as holding the door.

Quorum / unfireable mouth: LIVE needs more than one friendly operator vibe - policy quorum + CHARGE + Gate. Proof sits where the actor alone cannot forge it (Velaru verify).

First weld (named)

- Name: Withdraw / payout - clear before wire
- Prove DENY on one irreversible money-leave path
- Example path shape: POST /v1/payouts/{id}/release
- Human checkout door: /operator -> withdraw
- Next after prove: grid shed/reconnect. Nuclear C2 stays state-only.

Contribution ladder (altitude, not ownership fantasies)

- 1. Commercial driver weld (payout clear) - prove DENY
- 2. Critical infra mouth (grid) after prove
- 3. Named in coordinator plans + cleared custodians
- 4. Defense/comms release mouth under programs - unclaimed until real

Retrieval drill

- Explain can vs may with one money example.
- What is the first weld and why not grid first?
- What does fail-closed mean without using the phrase fail-closed?
- What do we refuse to own?